

Requirements

Project Task Tracker

Business Scenario

A software development team needs an application to track project tasks assigned to Employees.

Functional Requirements

1. Create a Login Page (username & pwd)
2. Add Employee
3. Search and List the Employees
4. Create Task
5. Search and Assign Task
6. Update Task Status

Employee Details

Employee ID, Employee Name, Department

Task Details

Task ID, Task Name, Priority (High/Medium/Low), Due Date, Status (Open/In Progress/Completed)

Assignment Rules

Employee and Task must exist. A task can be assigned to only one employee.

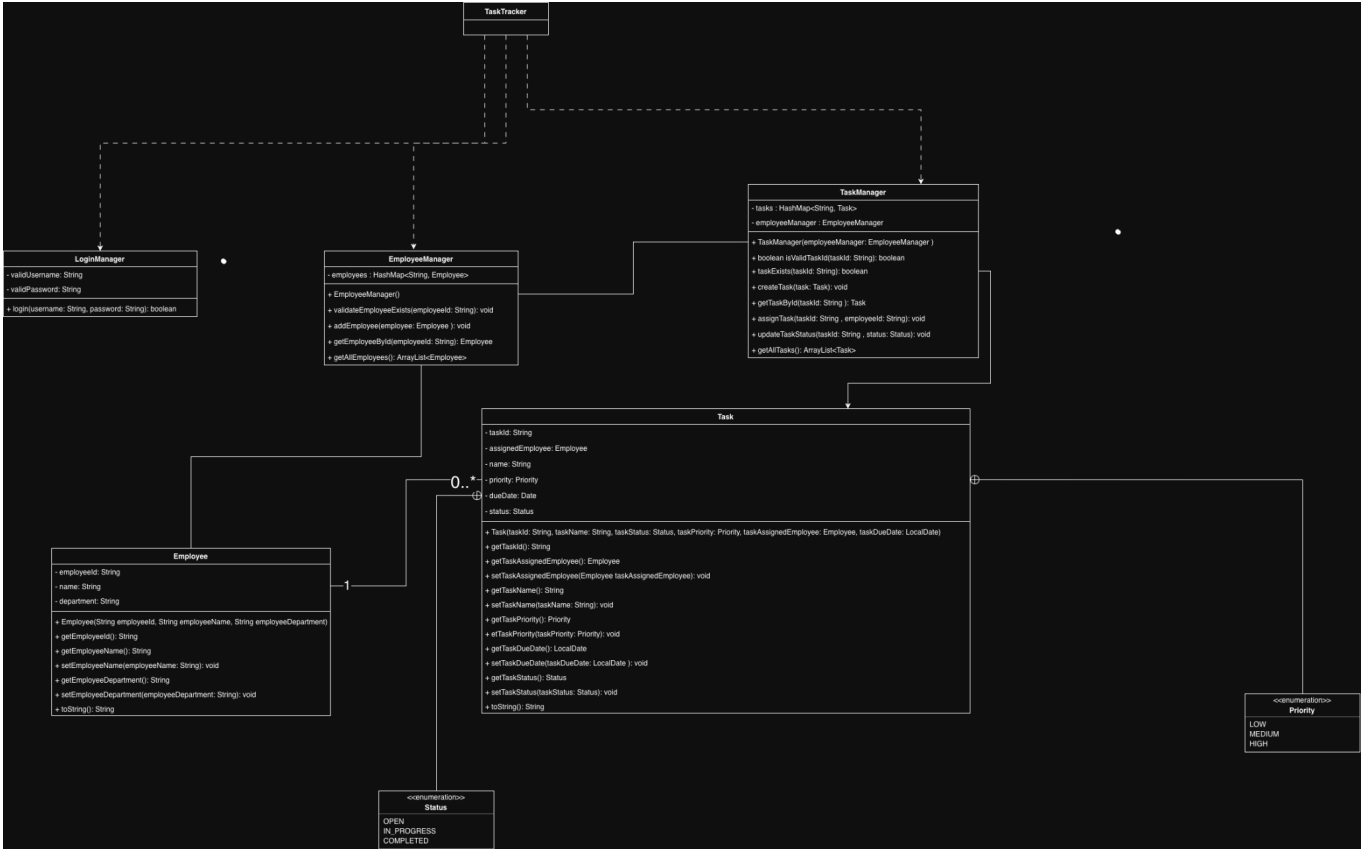
Technical Expectations

Use Java OOP concepts, HashMap, ArrayList, Exception Handling and MySQL table Design.

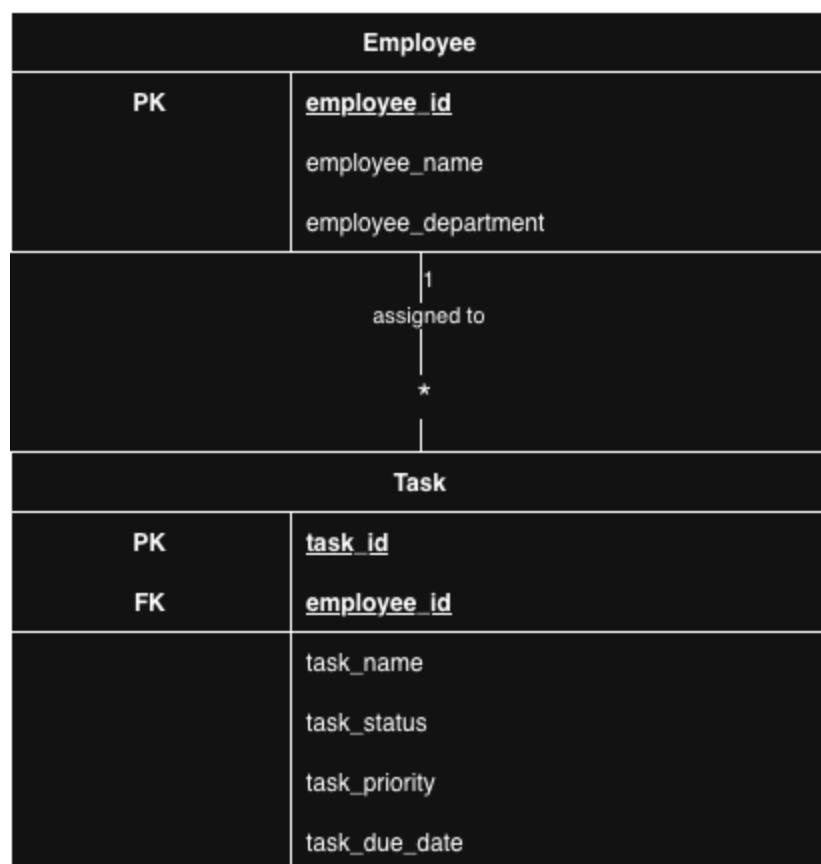
Exceptions

Employee Not Found, Task Not Found, Duplicate Task ID, Invalid Status

Class Diagram



MySQL Design (ERD)



Package Structure

```
src
├── main
│   ├── java
│   │   ├── org
│   │   │   └── tasktracker
│   │   │       ├── model
│   │   │       │   ├── Employee.java
│   │   │       │   ├── Task.java
│   │   │       │   ├── Priority.java
│   │   │       │   └── Status.java
│   │   │       ├── manager
│   │   │       │   ├── EmployeeManager.java
│   │   │       │   ├── TaskManager.java
│   │   │       │   └── LoginManager.java
│   │   │       ├── exception
│   │   │       │   ├── EmployeeNotFoundException.java
│   │   │       │   ├── TaskNotFoundException.java
│   │   │       │   ├── DuplicateTaskIdException.java
│   │   │       │   └── InvalidStatusException.java
│   │   │       └── TaskTracker.java
```


Positive Scenarios

Objective

To verify that an administrator can successfully add an employee, create a task, assign the task to an employee, and update the task status.

Preconditions

- The application is running.
- The user has valid login credentials.

Test Data

Item	Value
Username	admin
Password	admin123
Employee ID	E001
Employee Name	Mark
Department	IT
Task ID	AB-123
Task Name	Refactoring
Priority	LOW
Due Date	01-05-2026
Status	IN_PROGRESS

Test Steps

Step	Action	Expected Result
1	Log in using valid credentials.	User logs in successfully and the Main Menu is displayed.
2	Select Add Employee and enter valid employee information.	Employee is added successfully.

3	Select Create Task and enter valid task details.	Task is created successfully.
4	Select Search & Assign Task and assign the task to Employee E001.	Task is successfully assigned to Mark and the updated task details are displayed.
5	Select Update Task Status and change the status to IN_PROGRESS .	Task status is updated successfully and the updated task details are displayed.
6	Exit the application.	Application terminates successfully.

Actual Result

The employee was successfully created, the task was created, assigned to the selected employee, and the task status was updated successfully. The application displayed the updated task details after each successful operation.

Status

Pass

1. LOGIN Successful

```
=====
                        PROJECT TASK TRACKER
=====
Please log in to continue.

Username (or type 'exit'): admin
Password: password123

=====
[SUCCESS] Login successful!
Welcome, admin!
=====

=====
                        MAIN MENU
=====
1. Add Employee
2. Search/List Employees
3. Create Task
4. Search & Assign Task
5. Update Task Status
6. Exit
=====
```

2. Employee created

```
=====
                        MAIN MENU
=====
1. Add Employee
2. Search/List Employees
3. Create Task
4. Search & Assign Task
5. Update Task Status
6. Exit
=====
Select an option: 1

===== ADD EMPLOYEE =====
Employee ID: E001
Employee Name: Mark
Department: ICT

=====
[SUCCESS] Employee added successfully!
=====
```

3. Employee searched

```
=====
                        MAIN MENU
=====
1. Add Employee
2. Search/List Employees
3. Create Task
4. Search & Assign Task
5. Update Task Status
6. Exit
=====
Select an option: 2

===== SEARCH EMPLOYEE =====

Available Employees:
E001 - Mark

Employee ID (leave blank to list all): E001

===== EMPLOYEE DETAILS =====
Employee ID: E001
Employee Name: Mark
Department: ICT
```

4. Task created

```
=====
                        MAIN MENU
=====
1. Add Employee
2. Search/List Employees
3. Create Task
4. Search & Assign Task
5. Update Task Status
6. Exit
=====
Select an option: 3

===== CREATE TASK =====
Task ID (e.g. AB-123): AB-123
Task Name: Refactoring
Priority (LOW, MEDIUM, HIGH): LOW
Due Date (DD-MM-YYYY, e.g. 01-05-2026): 01-05-2026

=====
[SUCCESS] Task created successfully!
=====
```

5. Task searched and assigned

```

=====
                        MAIN MENU
=====
1. Add Employee
2. Search/List Employees
3. Create Task
4. Search & Assign Task
5. Update Task Status
6. Exit
=====
Select an option: 4

===== ASSIGN TASK =====

Available Tasks:
AB-123 - Refactoring (2026-05-01)

Task ID (or 0 to cancel): AB-123

===== TASK DETAILS =====
Task ID: AB-123
Task Name: Refactoring
Status: OPEN
Priority: LOW
Assigned Employee: None
Due Date: 01-05-2026

Available Employees:
E001 - Mark

Employee ID (or 0 to cancel): E001

=====
[SUCCESS] Task assigned successfully!
=====

===== UPDATED TASK DETAILS =====
Task ID: AB-123
Task Name: Refactoring
Status: OPEN
Priority: LOW
Assigned Employee: Mark
Due Date: 01-05-2026

```

6. Task status is updated


```

=====
MAIN MENU
=====
1. Add Employee
2. Search/List Employees
3. Create Task
4. Search & Assign Task
5. Update Task Status
6. Exit
=====
Select an option: 5

===== UPDATE TASK STATUS =====

Available Tasks:
AB-123 - Refactoring (2026-05-01)

Task ID (or 0 to cancel): AB-123
New Status (OPEN, IN_PROGRESS, COMPLETED): IN_PROGRESS

=====
[SUCCESS] Task status updated successfully!
=====

===== UPDATED TASK DETAILS =====
Task ID: AB-123
Task Name: Refactoring
Status: IN_PROGRESS
Priority: LOW
Assigned Employee: Mark
Due Date: 01-05-2026

```

7. User logged out

```

=====
MAIN MENU
=====
1. Add Employee
2. Search/List Employees
3. Create Task
4. Search & Assign Task
5. Update Task Status
6. Exit
=====
Select an option: 6

=====
You have been logged out.
Goodbye!
=====

```

Exception Scenarios

1. Employee ID not found

```
=====
                        MAIN MENU
=====
1. Add Employee
2. Search/List Employees
3. Create Task
4. Search & Assign Task
5. Update Task Status
6. Exit
=====
Select an option: 2

===== SEARCH EMPLOYEE =====

Available Employees:
E001 – Mark

Employee ID (leave blank to list all): E002

=====
[ERROR] Employee with ID E002 does not exist
=====
```

2. Task not found

```

=====
                        MAIN MENU
=====
1. Add Employee
2. Search/List Employees
3. Create Task
4. Search & Assign Task
5. Update Task Status
6. Exit
=====
Select an option: 4

===== ASSIGN TASK =====

Available Tasks:
AB-123 - Refactoring (2026-05-01)

Task ID (or 0 to cancel): AB-124

=====
[ERROR] Task with ID AB-124 does not exist
=====

```

3. Duplicate Task ID

```

=====
                        MAIN MENU
=====
1. Add Employee
2. Search/List Employees
3. Create Task
4. Search & Assign Task
5. Update Task Status
6. Exit
=====
Select an option: 3

===== CREATE TASK =====
Task ID (e.g. AB-123): AB-123
Task Name: Add a feature
Priority (LOW, MEDIUM, HIGH): LOW
Due Date (DD-MM-YYYY, e.g. 01-05-2026): 01-05-2026

=====
[ERROR] Task with ID AB-123 already exists.
=====

```

4. Invalid Status

```
=====
                        MAIN MENU
=====
1. Add Employee
2. Search/List Employees
3. Create Task
4. Search & Assign Task
5. Update Task Status
6. Exit
=====
Select an option: 5

===== UPDATE TASK STATUS =====

Available Tasks:
AB-123 - Refactoring (2026-05-01)

Task ID (or 0 to cancel): AB-123
New Status (OPEN, IN_PROGRESS, COMPLETED): a

=====
[ERROR] Invalid status. Valid statuses are: OPEN, IN_PROGRESS, COMPLETED.
=====
New Status (OPEN, IN_PROGRESS, COMPLETED): IN_PROGRESS

=====
[SUCCESS] Task status updated successfully!
=====

===== UPDATED TASK DETAILS =====
Task ID: AB-123
Task Name: Refactoring
Status: IN_PROGRESS
Priority: LOW
Assigned Employee: None
Due Date: 01-05-2026
```